

Project Summary

The objective of this project was to implement a Random Forest classifier entirely from scratch to predict passenger survival on the Titanic dataset. Unlike conventional implementations that rely on machine learning libraries such as scikit-learn, every component of the Decision Tree and Random Forest algorithms was manually developed using only Python, NumPy, and Pandas. This project followed the complete machine learning workflow, including exploratory data analysis, data preprocessing, feature engineering, model implementation, hyperparameter tuning, evaluation, and feature importance analysis.

A major focus of the project was understanding the underlying algorithms rather than treating machine learning models as black boxes. The Decision Tree implementation included recursive tree construction, Gini Impurity, Information Gain, random feature selection, and prediction through tree traversal. These trees were then combined into a Random Forest using bootstrap sampling and majority voting to improve prediction accuracy and stability.

Key Findings and Insights

Several observations from exploratory data analysis were reflected in the final model. Female passengers and first-class passengers had significantly higher survival rates, while higher fares were generally associated with improved survival. The dataset also contained substantial missing values, particularly in the **Cabin** feature, which contained over 75% missing data and was therefore removed to avoid introducing unreliable estimates.

Rather than relying on simple mean or mode imputation, a **hierarchical group-based imputation strategy** was designed for the remaining missing values. Missing **Age**, **Fare**, and **Embarked** values were estimated using passengers with similar characteristics, gradually relaxing the grouping criteria until a sufficiently large group was found. A separate tie-resolution function was also implemented for categorical features, allowing tied modes to be resolved using progressively broader groups. This preserved passenger-specific information while maintaining consistency across the preprocessing pipeline.

The project also introduced meaningful feature engineering. Passenger titles were extracted from names and grouped into common categories before one-hot encoding, while **FamilySize** was created by combining **SibSp** and **Parch**. Continuous variables such as **Age** and **Fare** were intentionally retained rather than binned, allowing the Decision Tree to determine optimal split thresholds during training instead of relying on manually defined intervals.

Feature importance analysis showed that **Title_Mr**, **Sex**, **Fare**, and **Passenger Class** were the strongest predictors of survival, closely matching patterns observed during exploratory data analysis.

Model Performance Summary

Hyperparameter tuning evaluated multiple combinations of the number of trees, maximum tree depth, and minimum samples required for splitting. Each configuration was trained and evaluated **10 times** using an 80–20 train-test split, with the average prediction accuracy and standard deviation used to compare performance. Although the highest average accuracy was achieved using 200 trees, the improvement over a 100-tree model was negligible. Consequently, a configuration of **100 trees, maximum depth of 5, and minimum samples split of 2** was selected because it provided the best balance between predictive performance and computational efficiency.

The final model was evaluated over **20 independent runs**, producing an average **accuracy of 87.57%** with a standard deviation of **0.61%**, indicating stable and consistent performance. It achieved a **Precision of 91.25%**, **Recall of 72.19%**, and an **F1-score of 80.58%**. The model demonstrated high reliability when predicting survivors while maintaining strong overall classification performance despite being implemented entirely without external machine learning libraries.

Challenges Faced and Solutions

The two most significant challenges during development were handling missing data and selecting appropriate hyperparameters.

Designing an effective missing-value strategy required balancing data preservation with reliability. Instead of applying simple global statistics, a hierarchical similarity-based imputation approach was developed, allowing missing values to be estimated using passengers with comparable characteristics. This required implementing reusable search and tie-resolution functions that could be applied consistently across multiple features.

Hyperparameter tuning was also computationally demanding because the Random Forest implementation was built entirely from scratch. To obtain reliable results, each parameter combination was trained and evaluated multiple times before comparing average performance and consistency. This process ensured that the final model was selected based on both predictive accuracy and robustness rather than a single experimental run.

Future Improvement Suggestions

Although the implemented Random Forest closely follows the standard algorithm, several enhancements were identified during development. One proposed modification is **dynamic random feature selection**, where each node randomly evaluates a variable number of candidate features instead of always using $\sqrt{p}$ features. Another proposed improvement is **variable bootstrap sample sizes**, allowing different trees to train on differently sized bootstrap datasets to further increase ensemble diversity.

Future work could compare these experimental modifications with the standard Random Forest implementation using identical training and testing splits to determine whether the additional randomness improves generalisation while maintaining model stability.